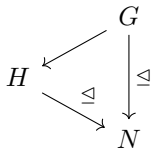
Topics in Algebra - Lecture 2

Given $N \trianglelefteq G$, what is the relationship between subgroups of G and subgroups of G/N?



We claim that $H/N < G/N$.
Proof: $\gamma_N : G \to G/N$ is defined by $\gamma_N(g) = gN$, we call this the canonical map. Using the fact that γ_N is a homomorphism, we have that

$$\gamma_N(H) < G/N \iff H/N < G/N$$

(Exercise: if H is normal in G, prove $H/N \trianglelefteq G/N$).
Recall: if $f : X \to Y$ is a function and $W \subseteq Y$, $f^{-1}(W) = \{x \in X \mid f(x) \in W\}$
Theorem 15.16: If $\phi : G \to G'$ is a homomorphism, and $N' > G'$ then
$\phi^{-1}(N') < G$
Lemma 34.3 (fraleigh): TODO
Proof: TODO

Example

